AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

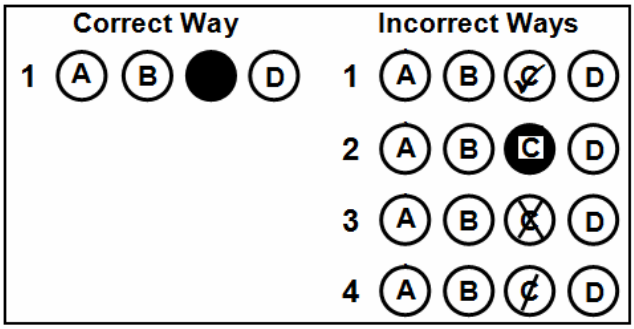
MAY 2015

General Mathematics Paper I

Time: 40 minutes Marks: 30

INSTRUCTIONS

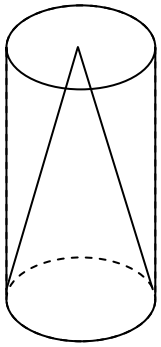
1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 30 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a simple calculator if you wish.

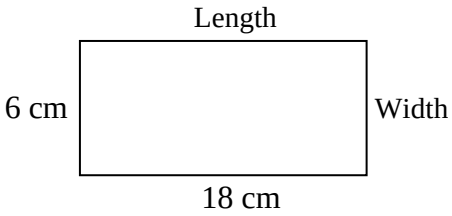
1. For equal radii, the ratio of volumes of a cone and a cylinder is 1 : 3. If volume of the cylinder is $6\ cm^3$, then the volume of cone of same radius is



A. $2\ cm^3$
B. $9\ cm^3$
C. $18\ cm^3$
D. $27\ cm^3$

2. The length and width of a rectangle are shown in the figure. The ratio of width to length is

A. 1 : 4
B. 1 : 3
C. 3 : 1
D. 4 : 1



3. If 8 workers can paint a house in 24 days, 18 workers can paint the same house in

A. 6 days.

B. $10\frac{1}{3}$ days.

C. $10\frac{2}{3}$ days.

D. 54 days.

4. Zakat payable on a man who has an annual saving of Rs 150,000 is

[Note: Rate of Zakat is 2.5% of annual saving.]

A. Rs 2,250
B. Rs 3,000
C. Rs 3,750
D. Rs 5,250

5. What will be the amount of Usher due on a cotton yield of Rs 265,000?

[Note: Usher is 10% of the yield.]

A. Rs 26.5
B. Rs 265
C. Rs 2,650
D. Rs 26,500

6. Sara sold her furniture for Rs 18,000 at a loss of 20%. The original price of her furniture was

   A. Rs $15,000$
   B. Rs $18,367$
   C. Rs $21,600$
   D. Rs $22,500$

7. A book seller bought 20 books for Rs 1,000. Fifteen of these books were sold for Rs 70 each and the rest for Rs 90 each. The profit gained is

   A. Rs 500
   B. Rs 750
   C. Rs 850
   D. Rs 1,250

8. A radio costs Rs 1,200. Its price in a 10% sale is

   A. Rs 1,080
   B. Rs 1,100
   C. Rs 1,188
   D. Rs 1,190

9. Alia pays 75% down payment on a property she bought for Rs x. Assuming there is no interest, the remaining amount is paid in 5 equal monthly installments of

   A. Rs $\frac{3x}{20}$

   B. Rs $\frac{x}{20}$

   C. Rs $5x$

   D. Rs $15x$

10. If 1 Rupee = $\frac{1}{115}$ Euro, then 11 Euro is equal to

    A. Rs 0.10
    B. Rs 10.45
    C. Rs 115
    D. Rs 1,256

11. The fixed rate of insurance of a car that costs Rs 1,300,000 is 30% per year. Due to an yearly 10% rate of depreciation, the value of the car is decreased by Rs 130,000 in the second year of its purchase. The insurance paid in the second year is

    A. Rs 351,000
    B. Rs 9,100
    C. Rs 26,000
    D. Rs 126,100

12. If the principal amount is Rs 2,000 and the rate of interest is 2% per annum, then after 1 year the markup will be

A. Rs 20
B. Rs 40
C. Rs 200
D. Rs 400

13. When the annual income exceeds Rs 400,000, 5% income tax is applicable on the exceeding amount. The annual income tax of a person having an annual salary of Rs 600,000 is

A. Rs 10,000
B. Rs 20,000
C. Rs 30,000
D. Rs 40,000

14. According to the table, total selling price of item *a* and *b* is

Item	Marked Price (Rs)	Sales Tax
a	1,000	10%
b	1,500	5%

A. Rs 2,500
B. Rs 2,575
C. Rs 2,675
D. Rs 2,875

15. The total salary of a person is Rs 20,000. After the deductions given in the table, his net (take home) salary is

Income tax	Rs 800
Provident fund	Rs 1,200
Life insurance	Rs 500

A. Rs 17,500
B. Rs 18,500
C. Rs 19,500
D. Rs 22,500

16. In an electric bill, the previous metre reading was 923452 units and current reading is 924232 units. If energy charges are Rs 7 per units excluding all other charges, then the total amount of energy charges is

A. Rs 4,780
B. Rs 4,860
C. Rs 5,280
D. Rs 5,460

17. $\left(\frac{2}{3}\right)^{-\frac{1}{2}}$ is equal to

A. $\sqrt{-\frac{2}{3}}$

B. $\sqrt{-\frac{3}{2}}$

C. $\sqrt{\frac{2}{3}}$

D. $\sqrt{\frac{3}{2}}$

18. The exponent in $a^{d} = (b+c)$ is

A. a
B. b
C. c
D. d

19. Log of a number x is 1.225. The characteristic of x is

A. 1
B. 3
C. 0.225
D. 0.005

20. If $T_3 = 2$ and $T_4 = 3$ in a geometric progression, where T_3 is the third term and T_4 is the fourth term, then the common ratio is equal to

A. $-\frac{3}{2}$

B. $-\frac{2}{3}$

C. $\frac{3}{2}$

D. $\frac{2}{3}$

21. The general term (n^{th} term) of the sequence $20, 13, 6, -1, -8,$ where n = 1, 2, 3 … is

A. $13 - 7n$
B. $27 - 7n$
C. $13 + 7n$
D. $27 + 7n$

22. The geometric mean between $\frac{4}{9}$ and $\frac{49}{64}$ is

A. $\frac{11}{20}$

B. $\frac{7}{12}$

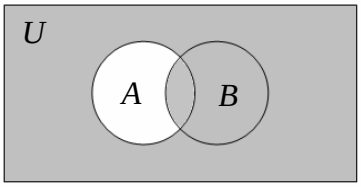
C. $\frac{16}{21}$

D. $\frac{697}{1152}$

23. $(B \cup C)^c$ is equal to

A. $B \cup C^c$
B. $B^c \cup C^c$
C. $(B \cap C)^c$
D. $B^c \cap C^c$

24. The shaded region in the given Venn diagram represents



A. A^c
B. B^c
C. $(A - B)^c$
D. $(B - A)^c$

25. The set of real numbers between 1 and 3 is a

A. null set.
B. finite set.
C. infinite set.
D. set containing one element.

26. Which of the following represents associative property of intersection of set *A*, *B* and *C*?

A. $A \cup (B \cup C) = (A \cup B) \cup C$

B. $A \cap (B \cap C) = (A \cap B) \cap C$

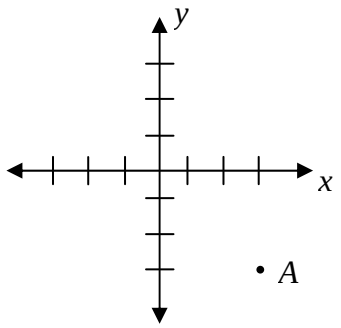
C. $A \cup (B \cup C) = (B \cup C) \cup A$

D. $A \cap (B \cap C) = (B \cap C) \cap A$

27. If $(0,\ k+1) = (0,-1),$ then the value of *k* is

A. -4
B. -2
C. -1
D. -3

28. In the given diagram, point *A* is



A. $(3,3)$

B. $(-3,3)$

C. $(3,-3)$

D. $(-3,-3)$

29. The average of 6 numbers is 10. If one of the numbers is 5, then what is the average of the remaining 5 numbers?

A. 10
B. 11
C. 12
D. 13

30. According to the following table, how many students obtained more than 30 marks?

Marks Obtained	Number of Students
11 – 20	3
21 – 30	7
31 – 40	15
41 – 50	8

A. 8
B. 10
C. 15
D. 23

END OF PAPER

Please use this page for rough work